

MCF3M: M5C2

Today:

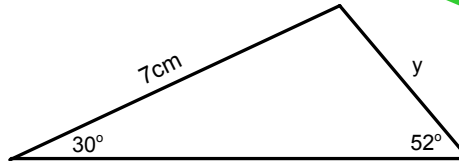
1. Minds-On:
2. Action: Using the Sine & Cosine Law
3. Consolidation: Exit Card

Learning Goal:

I will find angles and sides using the Sine Law & Cosine Law

Minds-On:

Consider the following triangle:

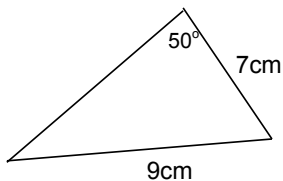


Could we use SOH CAH TOA to solve this?

The Sine Law:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \quad \text{OR} \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

- Used in NON right-triangles
- Use it when you know any pair of ***angle and opposite side***,



Or more specifically, if you know

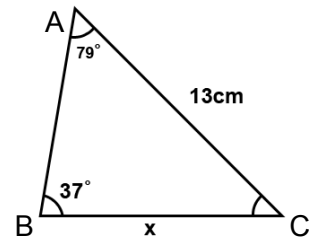
- ASA: Angle-Side-Angle
 - AAS: Angle-Angle-Side
 - ASS: Angle-Side-Side
- Only use 2 of the 3 fractions

Ex. 1: Use the Sine Law to determine x:

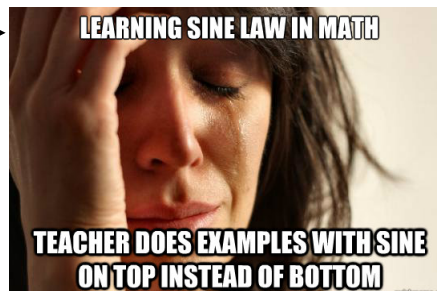
Step 1:

Step 2:

Step 3:



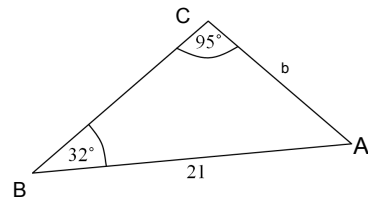
So why is this
supposed to be
funny?



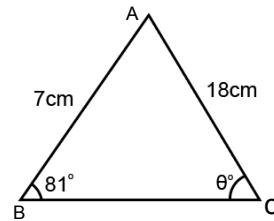
$$\frac{\sin 79^\circ}{x} = \frac{\sin 37^\circ}{13}$$

You Try It:

GoFormative M5C2: Sine Law Question 1



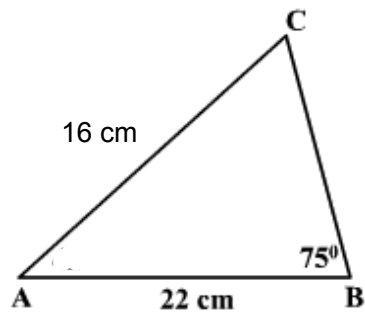
Ex. 2: Determine the measure of angle C:



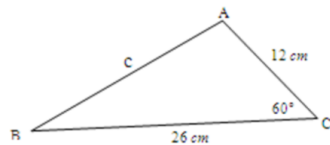
You Try It:

GoFormative M5C2: Sine Law Question 2

Determine the measure of angle C:



When dealing with non-right angle triangles, sometimes there is not enough information to use the Sine Law. For example, we cannot solve for c in the triangle below by using the Sine Law:



Fortunately, there is another law that exists that will allow us to calculate missing sides and angles for non-right angle triangles. It is called the Cosine Law

Cosine Law:

For sides:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

For angles:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

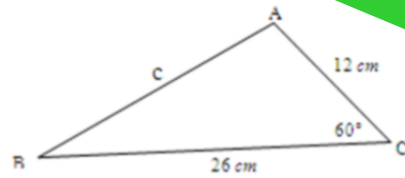
The cosine law can be used for SSS and SAS triangles

Note: If you're solving for a value other than A/a , you can either

- Re-label the triangle, or
- Adjust the formula:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

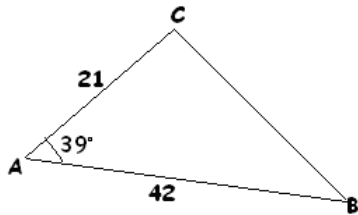
Ex. 3: Determine the length of side c :



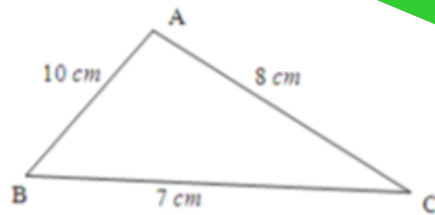
You Try It:

GoFormative M5C2: Cosine Law Question 1

Determine the measure of side a to one decimal place:



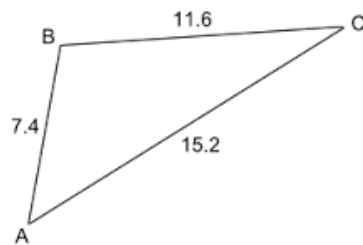
Ex. 4: Determine the measure of angle A:



You Try It:

GoFormative M5C2: Cosine Law Question 2

Determine the measure of angle B:



Consolidation:

Message me explaining how you determine if the triangle below could be solved using the sine or the cosine law.

